

Homework 8

Exercises

1. Let F be a field (Axioms A1–A8, A12–A13) and let $a, b \in F$.

(a) Suppose that $ab = 0$. Prove that $a = 0$ or $b = 0$.

Note: The proof here is different than it is for ordered rings. That said, a contrapositive argument is still a good idea! Note that the Cancellation Law is also easier to prove for fields.

(b) Suppose that $a^2 = b^2$. Prove that $a = b$ or $a = -b$.

(c) Conclude that if $x \in \mathbb{R}^+$, there is a unique $y \in \mathbb{R}^+$ such that $x = y^2$. (Given Theorem 6.14 in the notes)

All told, you have shown that $\sqrt{} : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is a well-defined function!

2. Let S be a non-empty subset of $\mathbb{R}$ that is bounded below. Prove that S has a **greatest lower bound**.

*Henceforth, we refer to this quantity as the **infimum** of S , denoted $\inf S$.*

3. Let $a, b \in \mathbb{R}$ with $a < b$ and consider the half-open interval $S = [a, b)$.

(a) Prove that $a = \inf S$.

(b) Prove that $b = \sup S$.

4. Let $S = \{\frac{1}{n} : n \in \mathbb{Z}^+\}$. Compute $\inf S$ and $\sup S$ and prove your answers.

5. Recall that $\alpha \in \mathbb{R}$ is called **irrational** if $\alpha \notin \mathbb{Q}$. Show that, if α is irrational, then so is $r\alpha$ for every $r \in \mathbb{Q} \setminus \{0\}$.

6. We reflect here on the **density** of $\mathbb{Q} \subset \mathbb{R}$.

(a) Prove that between any two distinct real numbers there is an *irrational* number.

(b) Prove that between any two distinct real numbers there are infinitely many rational numbers.

7. The following real numbers are given in ternary form. Express the number in base 10. Which are in the Cantor set $\mathcal{C}$?

(a) $x = 0.002002002 \dots$ (with 002 repeated indefinitely).

(b) $y = 0.011011011 \dots$ (with 011 repeated indefinitely).

(c) $z = 0.001000000 \dots$ (with 0's repeated indefinitely).

References